

NovaBank Banking Transaction Analysis

Full Project Report

Prepared for: Chief Customer Officer / NovaBank

1. Business Context & Objective

NovaBank is a mid-sized retail bank operating across Maharashtra and Karnataka, serving approximately 85,000 retail customers through branch and digital channels.

For this analysis, a representative sample of 297,129 transactions, 5,000 customers, 6,356 accounts, 10 branches was analyzed to understand customer engagement, retention, and transaction behavior.

The Chief Customer Officer observed that customer engagement and retention were not improving despite increasing transaction activity. The objective of this project was to identify the root causes behind declining engagement and provide data-driven recommendations to improve customer activation, retention, and long-term customer value.

Tools Used: SQL (Data Cleaning), Python (Feature Engineering & Analysis), Tableau (Dashboards)

2. KPI Framework

Customer Retention Rate was treated as the primary KPI, supported by activation, engagement, transaction reliability, and financial behavior metrics.

Key supporting indicators included:

- Never Transacted Customers
 - Transaction Success Rate
 - Monthly Active Transaction Users (MATU)
 - RFM Segmentation
 - Cash Inflow vs Outflow
 - Customer Lifecycle Stage
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3. Data Understanding & Preparation

The project followed a normalized relational schema with four tables: customers, accounts, transactions, and branches.

Key data preparation steps included:

- Investigated 46 duplicate customer emails and retained records due to linked account activity.
- Imputed missing transaction_type and is_online values using business logic.
- Flagged 2,721 non-transacting customers and 3,414 non-transacting accounts.
- Excluded unreliable balance values caused by data-generation artifacts and used inflow/outflow ratio as an alternative behavioral signal.

Feature engineering was performed in Python and included:

- RFM Scoring & Segmentation
 - Lifecycle Stage Classification
 - Days Since Last Transaction
 - Balance Trend Analysis
 - Cohort Retention Metrics
 - Monthly Active Transaction Users (MATU)
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4. Analysis & Key Findings

Customer Analysis

- Retention Rate: 40.59% vs retail banking benchmark of 75–85%.
- 54.42% of customers (2,721 out of 5,000) never completed a single transaction after onboarding.
- 94.3% of customers fall into Low Engagement segments, while only 18 customers qualify as High Value based on RFM segment.

Key Finding:

NovaBank's biggest challenge is customer activation, not customer loyalty. Customers who activate early retain strongly, but most customers never begin transacting.

Transaction Analysis

- Average transaction value: ₹8,240.
- Transaction Success Rate: 82.68%, meaning nearly 1 in 6 transactions fails.
- 85.1% of transactions are digital, confirming strong digital adoption.
- Monthly Active Transacting Users increased from 22 users in January 2022 to 1,849 users in November 2023 (84× growth).

Key Finding:

NovaBank has successfully driven digital adoption, but increased activity has not translated into stronger retention.

Financial & Behavioral Analysis

- Total portfolio outflow exceeds inflow by ₹2 Crore.
- 99.4% of balances sit in active accounts, indicating customers are financially capable.
- 310 customers show “Inflow Only” behavior, receiving money but spending elsewhere.
- No strong festive season spending spike was observed, suggesting NovaBank behaves like a secondary bank for many customers.

Key Finding:

The disengagement problem is behavioral rather than financial.

Cohort Analysis

- Average Month 0 activation rate: 40.15%.
- Customers who activate in Month 1 retain at 65–91% for 12+ months.
- 2023 cohorts show weaker Month 1 retention than 2022 cohorts, indicating deteriorating onboarding quality.

Key Finding:

Customers who start transacting early remain loyal. The challenge lies in encouraging first-month activation.

5. Dashboard Overview

Five Tableau dashboards were developed to communicate insights:

1. Executive Overview – Customer health, retention, transaction success, and cash flow KPIs.
2. Customer Journey & Engagement – Customer lifecycle, MATU growth, city distribution, and engagement behavior.
3. Retention & Cohort Analysis – Cohort heatmap and onboarding quality trends.
4. Transaction & Behavioral Analysis – Transaction trends, payment behavior, and digital adoption.
5. Financial & Account Analysis – Account activity, inflow/outflow patterns, and portfolio health.

6. Root Cause Analysis

What is happening?

NovaBank’s retention rate is only 40.59%, driven primarily by a 54.42% activation failure rate and weak customer engagement.

Why is it happening?

Three major drivers were identified:

- Weak onboarding with no strong first-transaction incentive.

- High transaction failure rate (16.5%) causing early customer friction.
- Limited recurring engagement, leading customers to use NovaBank as a secondary banking account.

What is NOT causing the issue?

- Customer financial capacity is not the problem, as most balances remain healthy.
 - Loyalty strategy is not the main issue — customers who activate retain strongly.
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7. Strategic Recommendations

R1 — Improve Customer Activation (High)

Launch first-transaction cashback and UPI reward campaigns within the first 30 days of onboarding.

Expected Impact: Improve Month 0 activation and long-term retention.

R2 — Reduce Transaction Failures (High)

Audit payment systems and reduce transaction failure rate below 5%.

Expected Impact: Better customer trust and reduced early disengagement.

R3 — Increase Product Engagement (Medium)

Introduce auto-pay, cashback offers, and engagement nudges for low-activity customers.

Expected Impact: Higher transaction frequency and lower dormancy.

R4 — Cross-Sell to Salaried Customers (Medium)

Offer pre-approved loans to customers with stable salary credits.

Expected Impact: Increased customer lifetime value.

R5 — Protect High Value Customers (High)

Provide premium servicing and relationship management to top-value customers.

Expected Impact: Improved portfolio stability and revenue protection.

8. Conclusion

NovaBank's challenge is not customer acquisition but customer activation. The analysis shows that customers who become active early retain strongly, while onboarding friction and transaction failures reduce engagement.

The highest-value opportunities for NovaBank lie in improving first-month activation, reducing transaction failures, and increasing recurring engagement. By acting on these drivers, NovaBank can improve customer retention and long-term portfolio value.

9. Limitations

The dataset was synthetically generated to simulate realistic retail banking behavior and should be treated as a directional analytical exercise.

Transaction data covers January 2022 to December 2023 only. Customers who joined late in 2023 have limited behavioral history, which may slightly understate activation and retention metrics for recent cohorts.